

1. Curriculum

23CSE301

MACHINE LEARNING

L-T-P-C: 3-0-2-4

1.1 Course Objectives

Machine learning covers significant ground in various verticals – including image recognition, medicine, cyber security, facial recognition, business analytics etc. Study of ML lays a strong foundation for the study on NLP, Deep Learning, Reinforcement Learning, Graphical Models, AI, Predictive Analytics etc.

- The aim of this course is to provide foundational knowledge in machine learning.
- The students will learn to implement, train and validate the machine learning models and understand the recent algorithms in machine learning through case studies.

Pre-Requisite(s): 23MATXXX Linear Algebra, 23MATXXX Probability and Random Processes

1.2 Course Outcomes

CO1: Understand the fundamental concepts, issues and challenges of machine learning.

CO2: Implement machine learning algorithms using programming tools and provide solution to real world applications.

CO3: Apply machine learning algorithms for parameter estimation or prediction problem.

CO4: Apply supervised learning techniques for classification problem and analyse their performance.

CO5: Apply un-supervised and ensemble learning techniques to solve a given problem.

1.3 CO-PO Mapping

PO/PSO	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	P012	PS01	PS02
CO														
CO1	3	2	2	3									3	2
CO2	3	2	3	2	3	1		3	3	3		3	3	3
CO3	3	2	2	2									3	3
CO4	3	2	2	2									3	3
CO5	3	2	2	2									3	3

1.4 Syllabus

Unit 1

Introduction to machine learning, Types of Machine learning, features, class boundary, Training, Validation & Testing, Generalization - underfit, regular fit and overfit; Loss /Cost function; Issues of Explainability, algorithmic bias, data and algorithmic privacy. Curse of Dimensionality, Dimensionality Reduction Techniques – Principal component analysis, Linear Discriminant Analysis, Feature selection – sequential & bi-directional. K-Nearest Neighbour classifier, Regression: Linear, logistic, introduction to Regularization- LASSO, Ridge. Classifier Performance metrics – precision, recall, accuracy, f-score, AUC, Regression Performance metrics –RMSE, MAPE, R2 Score.

Unit 2

Supervised learning: Decision Trees, Support vector machines, Naive Bayes, Markov model, Hidden Markov Model, Artificial Neural Network - Perceptron & MLP with learning algorithms, Parameter Estimation: MLE and Bayesian Estimate, Expectation Maximization.

Unit 3

Unsupervised learning- Clustering – hierarchical & agglomerative, K means algorithm, cluster evaluation – Elbow technique; Ensemble learning – Bagging and boosting- Adaboost, Random Forest. Introduction to Reinforcement learning. Implementation of machine learning algorithms on real world problems and analyze their performance by tuning hyper parameters.

Textbook(s)

Tom Mitchell, "Machine Learning". McGraw Hill; ISBN: 9781259096952, 9781259096952; 2017.

Reference(s)

Kevin P. Murphy. "Machine Learning", a probabilistic perspective. MIT Press, 2012.

Christopher M Bishop. "Pattern Recognition and Machine Learning". Springer 2010.

Richard O. Duda, Peter E. Hart, David G. Stork. "Pattern Classification". Wiley, Second Edition; 2007.

Trevor Hastie, Robert Tibshirani, Jerome Friedman. "The Elements of Statistical Learning: Data Mining, Inference, and Prediction". Springer, 2017.

1.5 Evaluation Pattern

Machine Learning Evaluation and Grading				
Internal (70)			External (30)	Total
Components		Marks	Total Marks	
Mid term		20	20	
Continuous Assessment (Theory)	Quiz	10	50	End Semester = 30 Internal + External = 100
	Lab Activities (6)	05		
	Assignments (3)	10		
	Phase - 1: Problem Statement Definition and Interpretation	05		
	Phase - 2: Features Selection & Learning Technique Identification	10		
	Phase - 3: End Sem Capstone Evaluation	10		
Continuous Assessment (Lab): Capstone Project				